

PROJECT AND TEAM INFORMATION

Project Title

GeoLocal-Miner: A Multi-Scale Spatial Mineral Prospectivity and Exploration Target Prioritization Framework

Student / Team Information

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PROPOSAL DESCRIPTION

Motivation

1. Mineral exploration often involves analysing information from multiple sources such as geological characteristics, geophysical observations, terrain and remote sensing. The challenge is that these datasets describe large geographical areas, while detailed field investigation of the entire area can be time-consuming and resource-intensive.
2. Atomic Minerals Directorate (AMD) investigates monazite-bearing beach and inland placer deposits, with monazite being an important source of thorium and rare earth elements in India. AMD's exploration involves geological, geomorphological, geophysical and sampling-based investigations.
3. Our project proposes a small-scale computational prototype that demonstrates how already available spatial information can be organized into a grid and used to prioritize locations for further investigation.
4. The main motivation is not to detect or prove the presence of thorium, but to answer:
5. Can multiple spatial indicators be combined into a simple computational scoring system to identify and rank areas that deserve further geological investigation?
6. The project is inspired by the idea of high-resolution spatial modelling used in other fields. For example, IMD operates high-resolution spatial modelling systems at resolutions such as 2 km. We take the general concept of representing a large region through spatial cells and adapting it to mineral prospectivity analysis—not the meteorological model itself.

State of the Art / Current solution (1 pt)

- (i). Government organizations such as AMD and GSI already perform sophisticated mineral exploration and mineral-prospectivity activities using geological, geophysical, remote-sensing and other exploration datasets. GSI's mineral-hunt initiatives explicitly involve integration of multiple geoscience datasets and generation of predictive exploration-target maps.
- (ii). AMD's exploration of beach-sand minerals includes geological and geomorphological investigation, sampling and drilling, and covers monazite-bearing deposits across several Indian states.
- (iii). Our project does not claim to replace these systems. Instead, it proposes a simplified educational prototype focused on local spatial prioritization.
- (iv). The proposed system will take selected existing data indicators, divide a small study region into spatial cells, calculate a rule-based prospectivity score for each cell, and rank the cells according to exploration priority.
- (v). Unlike large-scale professional systems, our prototype will deliberately avoid machine learning and complex geological modelling. Its contribution is the demonstration of how C++ OOP, Data Structures, spatial grids, algorithms and visualization can be combined into a working exploration decision-support prototype.

Project Goals and Milestones

The main goal of our project is to develop GeoLocal-Miner, a C++-based spatial mineral exploration prioritization system that uses available geological, geophysical/radiometric and terrain information to evaluate a selected study region. The region will be divided into spatial zones, and each zone will be assigned a prospectivity score using a rule-based weighted formula. The system will then rank the zones and identify areas that may require further geological or geophysical investigation. The project will demonstrate the practical use of Object-Oriented Programming, Data Structures along with a graphical user interface.

Major milestones of the project are:

- (i).Select a suitable mineral, study region and relevant dataset.*
- (ii).Design and create classes for zones, datasets and analysis using and OOPs concepts.*
- (iii).Implement the prospectivity scoring formula for evaluating each spatial zone.*
- Use vectors, graphs, queues, linked list and more data structure to process and rank the zones.*
- (iv).Implement the classification of zones into High, Medium, Low priority levels.*
- (v).Develop the GUI to display the study region, targeted regions, zone scores and priority rankings.*
- (vi).Compare the generated priority zones with available known mineral-occurrence information for basic validation.*
- (vii).Integrate and test the complete system using different datasets.*

Project Approach

We will divide the approach of this project into different parts so that each part can be developed as well as tested before merging everything

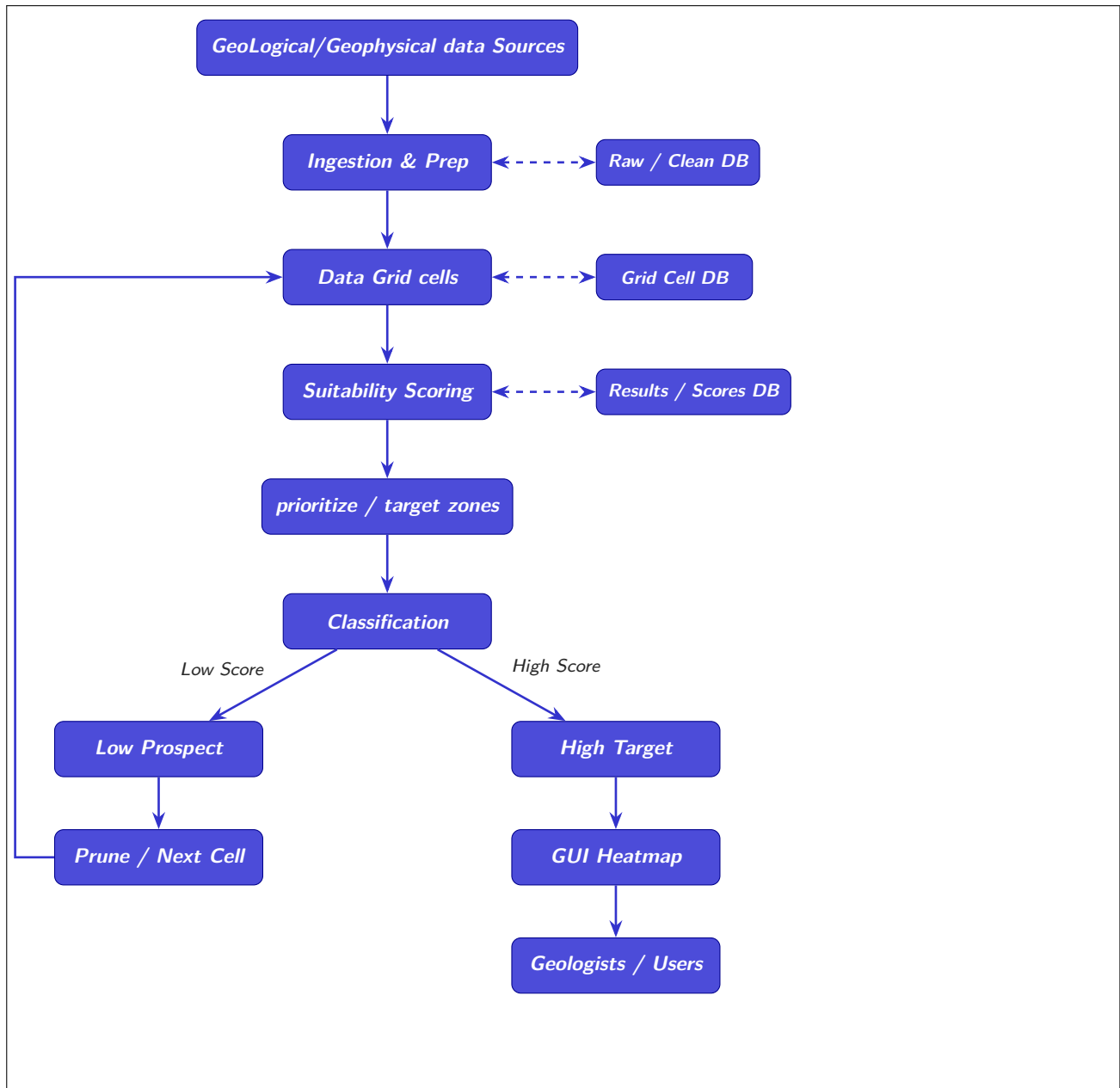
1. Data Preparation Grid: Select one mineral and a small study region, then collect available geological, geophysical/radiometric, terrain and known-occurrence data. Convert the information into a common grid where each cell contains relevant attributes and normalized scores.

2. C++ DS OOP Based Analysis: Develop the core system in C++ using OOPs classes such as Zone, Grid, DataLayer and PriorityEngine. Use vectors to store zones, graphs to represent neighbouring cells, and priority queues to rank areas. Calculate the prospectivity score using a formula based on the data

where G, R, S and T will be used in the formula in which they represent geological, radiometric, structural and terrain suitability respectively.

3. Visualisation Exploration Prioritisation: Build a graphical UI using ImGui via CMAKE to display the study region, prospectivity grid, scores and ranked priority zones. The system will classify areas as High, Medium, Low priority and recommend high-scoring zones for further geological/geophysical investigation. The results will be compared with available known occurrences for basic validation.

System Architecture (High Level Diagram)



Project Outcome / Deliverables (1 pts)

1. This project will produce a working C++ desktop prototype called GeoLocal-Miner.
2. The expected outputs are:
 - (i). A spatial grid representing the selected study region.
 - (ii). A prospectivity score for every spatial cell.
 - (iii). A ranked list of high-priority zones.
 - (iv). A visual prospectivity map/grid.
 - (v). Detailed information for a selected zone.
 - (vi). Implementation of Data Structures such as vectors, queues and graphs.
 - (vii). Demonstration of OOPs concept including classes, encapsulation, abstraction etc.

Assumptions

- (i).The selected study region has sufficient publicly accessible geological information for a prototype.*
- (ii).Geological and geophysical indicators can be represented as normalized scores for the prototype.*
- (iii).The initial weighting formula is a prototype scoring model, not a universally established geological law.*
- (iv).A high prospectivity score indicates priority for further investigation, not confirmed mineralization.*
- (v).Known mineral occurrences which are available, can be used as reference points for basic validation.*
- (vi).data quality will determine the achievable resolution.*

References

- (i).Atomic Minerals Directorate for Exploration and Research (AMD) — Beach Sand Minerals deposits Investigations. (<https://amd.gov.in/>)*
- (ii).Geological Survey of India (GSI) — Innovative Mineral Hunt Techniques / Mineral Prognostication. Demonstrates the current use of multi-source geoscience data and predictive mapping for mineral exploration.(<https://gsi.gov.in/home/>)*
- (iii).India Meteorological Department (IMD) — High Resolution Rapid Refresh (HRRR). Provides the methodological inspiration for high-resolution for prototype, IMD's HRRR operates at 2-km horizontal resolution at extreme local level.(https://mausam.imd.gov.in/index_n.php)*
- (iv).Refrence projects.(<https://github.com/>)*
- (v).Data structure and OOPs concepts tutorial.(<https://www.geeksforgeeks.org/>)*